



Review

Transforming plant-based waste and by-products into valuable products using various “Food Industry 4.0” enabling technologies: A literature review

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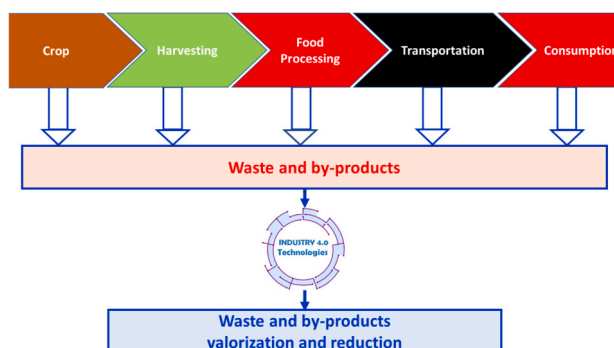
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HIGHLIGHTS

- Plant-based food production generates by-products rich in nutrients and bioactive compounds.
- Food Industry 4.0 technologies can transform plant-based waste or by-products into valuable products.
- Food Industry 4.0 technologies can contribute to higher sustainability and effective food system.

GRAPHICAL ABSTRACT



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ABSTRACT

The last several years have seen unprecedented strain on food systems as a result of pandemics, climate change, population growth, and urbanization. Thus, academic and scientific communities now view global food security as a critical issue. However, food loss and waste are a major challenge when adopting food security and sustainability strategies, since a large proportion of food is lost or wasted along the food supply chain. In order to use resources efficiently and enhance food security and sustainability, food waste and by-products must be reduced and properly valorized.

Plant-based food production generates various by-products which are generally rich in nutrients and bioactive compounds. Emerging technologies have been effectively employed to extract these valuable compounds with health benefits. Recently, Industry 4.0 technologies such as artificial intelligence, the Internet of Things,

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blockchain, robotics, smart sensors, 3D printing, and digital twins have a great deal of potential for waste reduction and by-products valorization in food industry. Reducing food waste not only benefits the environment, but also reduces greenhouse gas emissions and thus contributes to sustainable resource management. This review provides up-to-date information on the potential of Industry 4.0 for converting plant-based waste and by-products into valuable products.

Recent studies showed that innovations in Industry 4.0 provide attractive opportunities to increase the effectiveness of manufacturing operations and improve food quality, safety and traceability. By leveraging Food Industry 4.0, companies can transform plant-based waste and by-products into valuable products and contribute to a more sustainable and efficient food production system.

1. Introduction

Food systems have been under unprecedented challenges over the last few years due to climate change, human population growth, increased urbanization, and the outbreak of pandemics, as well as violence and conflicts in many parts of the world. Therefore, global food security has become a pressing issue and received much attention from scientific communities and academic institutions (Mehrabi et al., 2022; Tamasiga et al., 2023). Among the significant dilemmas that face implementing food security strategies and principles of food sustainability is food waste and loss, since a huge amount of food is lost or wasted along the food supply chain. Indeed, according to many references, one-third of all food produced is wasted globally, which corresponds to 1.3 billion tons of edible food each year (Mokrane et al., 2023; Roy et al., 2023; Stefanini and Vignali, 2023). Consequently, reducing and better valorizing food waste and by-products have been widely recommended as efficient strategies to maximize the benefits of available resources and improve food security, sustainability, and circularity in food systems (Aschemann-witzel et al., 2023; Rakesh and Mahendran, 2024; Thorsen et al., 2024).

Plant-derived waste and by-products, such as peel, leaves, seeds, bran, kernel, pomace, and oil cake can be obtained from a wide variety of fruit, vegetables, cereals, oil crops, root and tubers and other legumes during the primary production and the subsequent processing and distribution stages (Difonzo et al., 2021; Q. Jin et al., 2018; Pathania and Kaur, 2022). Growing research shows that such waste and by-products can be valorized by the extraction of valuable nutrients and compounds, such as proteins, vitamins, fatty acids, amino acids, dietary fibers, flavonoids, and phenolic compounds, among others (Ali et al., 2021; Hadidi et al., 2024; Joshi et al., 2022; Nirmal et al., 2023). The extracted components and bioactive compounds have been reported to have several health benefitting attributes to various health complications such as diabetes, cancer, and cardiovascular and neurocognitive diseases, mainly due to their antioxidant and antimicrobial activities (Perra et al., 2022; Szabo et al., 2022).

Although thermal treatments and other conventional methods have been widely used to recover high-added-value compounds from food waste and by-products, recent years have witnessed an accelerated development of green technologies and other emerging approaches. Unlike traditional methods, emerging techniques such as microwave, ultrasound, supercritical fluid, and pulsed electric field enable reduced processing time, improved yield, and better quality of the extracted compounds (Calleja-Gómez et al., 2024; Esparza et al., 2020; Venugopal et al., 2023).

Since the emergence of the fourth industrial revolution (called Industry 4.0), a growing emphasis has been placed on the convergence of physical, digital, and biological technological innovations (Hassoun et al., 2022b; Hassoun et al., 2023a, 2023b, 2023c, 2023d, 2024). The invention of steam engine and the emergence of the first factories during the first industrial revolution were followed by the arrival of electricity and the introduction of the concept of mass production during the second industrial revolution. The third industrial revolution began in the 1970s, through partial automation and the advent of the first computers and electronic devices, while the ongoing Industry 4.0 has been

characterized by the rapid development of advanced technological innovations such as artificial intelligence, big data, smart sensors, 3D printing, digital twins, robotics, blockchain, among others (Hassoun, 2024; Liebig et al., 2023). Due to their high potential and beneficial characteristics, the application of these technologies has increased in many agriculture and food related sectors.

From Fig. 1, it can be noticed that the number of publications and citations on the use of Industry 4.0 in food waste and loss has increased significantly over the last few years. Recent publications have shown that Industry 4.0 technologies offer significant opportunities in reducing and valorizing waste and by-products in the seafood (Hassoun et al., 2023c) and the dairy (Hassoun et al., 2024) sectors. In another recent publication, it was shown that harnessing Industry 4.0 technologies, especially artificial intelligence, big data, blockchain, smart sensors, 3D printing, as well as recent advances in nanobiotechnology could help reduce food waste, creating new food packaging solutions (Hassoun et al., 2023a, 2023b, 2023c, 2023d). Additionally, it was demonstrated in a recent study that Industry 4.0 could enable unlock the circularity of resources within food supply chains, leading to reduced food loss and waste (Lopes de Sousa Jabbour et al., 2021).

While our previous work and review publications explored the role of Industry 4.0 technologies in achieving sustainable development goals across the broader food sector, this review specifically focuses on the transformation of plant-based waste and by-products into valuable products. The work highlights the application of Food Industry 4.0 enabling technologies in waste valorization, offering a targeted analysis of innovative solutions for addressing waste management challenges within the plant-based food sector. This review presents novel insights and detailed discussions on specific technologies and methodologies that were not covered in the previous work, thereby contributing new knowledge to the field.

2. Overview of plant-based waste and by-products

Plant-based food production generates various by-products with valuable potential (Fig. 2).

Fruit and vegetable processing discards peels, stems, leaves, and seeds, which are often rich in fiber, alkaloids and vitamins (Nabi et al., 2024). Similarly, grain and cereal processing yields nutrient-rich by-products like bran and germ, containing protein, fiber, and micro-nutrients (Fărcaș et al., 2022). Oilseed processing, after extracting the main oil, generates meal or cake rich in protein, polyphenols, and residual oil (Sá et al., 2021). These readily accessible, inherently biodegradable plant by-products contribute to sustainable practices. Leveraging them through the recovery of valuable bioactive compounds presents a significant opportunity to contribute to the United Nations' 2030 Sustainable Development Goals (SDGs).

2.1. Fruits and vegetables by-products

Fruits and vegetables generate substantial quantities of waste, comprising stems, leaves, peels, and seeds (Tahir et al., 2023). Due to their inherent attributes, high quantities of waste and by-products from fruits and vegetables can be generated during the selection, washing,

cleaning, peeling, and nucleating procedures involved in their industrial processing. Furthermore, the presence of enzymes makes them susceptible to enzymatic browning, thus making waste generation an unavoidable outcome during the processing stage (Zhu et al., 2023). In developing countries, fruit and vegetable waste ranges from 15 % to 50 % of harvested produce and is typically generated before reaching the market (Gardas et al., 2017). Worldwide, approximately 0.3 and 15.6 million tons of kiwi residue and citrus peel waste are generated annually, respectively (Zhu et al., 2023). Furthermore, global grape pomace from grape processing accounts for 5–9 million tons annually (Zhu et al., 2023).

By-products from fruits and vegetables are rich in various beneficial compounds, including vitamins, carotenoids, alkaloids, proteins, dietary fibers, and polyphenols. Fruit waste, in particular, comprises cellulose, lignin, hemicellulose, and other polysaccharides (Nabi et al., 2024). Therefore, there is considerable potential to recover these natural components from such residues within the framework of the circular economy. The citrus industry generates around 50–60 % of by-products, totaling over 60 million tons globally (Wedamulla et al., 2022). These residual materials comprise peels, pulp, seeds, and leaves and are rich in bioactive compounds like polyphenols, carotenoids, pectin, essential oils, and enzymes. Orange peels are among the most prevalent waste products in the food industry and serve as a rich source of carotenoids. The carotenoid content in orange peels is estimated to be around 97.4 µg/g when acetone is employed as a solvent (Murador et al., 2019). An enhanced carotenoids extraction yield of 65,350 µg/g from orange peels was achieved using hydrophobic deep eutectic solvents (DES) consisting of Menthol and Camphor, along with an ultrasound extraction method with an ultrasound intensity of 60 % and an extraction time of 20 min (Viñas-Ospino et al., 2023).

Pomegranate juice processing plants produce a significant amount of waste, comprising fruit residues such as seeds and outer peels. The peel, consisting of the exocarp and mesocarp, accounts for approximately 50 % of the total weight of the fresh pomegranate fruit, and its chemical composition varies depending on cultivar type (Lacivita et al., 2023). This amounts to over 1.9 million metric tons per year globally (El Bar-nossi et al., 2021). The exceptional properties of pomegranate fruit peels (*Punica granatum* L.) are well-documented, as they contain a plethora of bioactive phytoconstituents, notably phenolic compounds like hydrolyzable ellagitannins, anthocyanins, and flavonoids (Singh et al., 2019). These compounds are renowned for their antioxidant, antimicrobial,

antifungal, antiviral, and antidiabetic activities (George et al., 2019; Lacivita et al., 2023).

Waste and by-products generated from tomatoes, the second most important vegetable crop Worldwide, are rich sources of carotenoids (lycopene, beta-carotene, lutein), flavonoids, phenolic acids, tannins, dietary fibers, and vitamin C (Khatib et al., 2023). These compounds contribute to the potential health benefits of tomato by-products, including antioxidant and anti-inflammatory effects, cardiovascular protection, and potential anticancer properties (Natali et al., 2023). Recently, Khatib et al. (2023) reported a polysaccharide yield of 6 % (w/w based on dry matter) for dried tomato by-products using water at 100 °C. They revealed that tomato polysaccharides are rich in rhamnose, with galacturonic acid content ranging from 22.3 % to 33.8 %. Additionally, they found that their degree of methylation (DM) and degree of acetylation (DA) varied from 8.86 % to 71.00 % and from 8.57 % to 19.42 %, respectively, depending on the solvent nature and temperature (Khatib et al., 2023). The observed variation in DA and DM levels could influence the functional and biological characteristics of polysaccharides recovered from plant by-products and waste, thereby determining their suitability for use in food and cosmetic industries. Polysaccharides extracted from the by-products of red bell peppers, a versatile vegetable cultivated and utilized globally, demonstrate notable anti-inflammatory, anti-complement, immunomodulatory, and chemopreventive properties (Adami et al., 2018). Additionally, these polysaccharides can be extracted at a yield nearing 10 % (Khatib et al., 2023). Red bell peppers also contain significant quantities of vitamin C, dietary fibers, flavonoids such as quercetin and luteolin, and carotenoids (Anaya-Esparza et al., 2021).

Dietary fibers, which form the structural framework of plant cell walls, comprise both soluble fibers, such as mucilage and pectin, and insoluble fibers, like lignin and cellulose. In lemon, orange, and grapefruit seeds, the content of soluble dietary fibers varied between 4.6 % and 7.9 %, while insoluble dietary fibers ranged from 75.9 % to 82.2 % (Karaman et al., 2017). The total dietary fibers in banana and mango peels were 64.3 % and 69.8 %, respectively (Tlais et al., 2020). Banana peels are mainly rich in insoluble fibers (56.8 %) like cellulose, hemicellulose, and lignin, but contains also substantial amounts of pectin (Arun et al., 2015). Dietary fiber consumption has been linked to numerous health advantages, such as a decreased risk of diabetes, cancer, hypertension, obesity, and cardiovascular and gastrointestinal disorders (Hasan et al., 2024; Tlais et al., 2020).

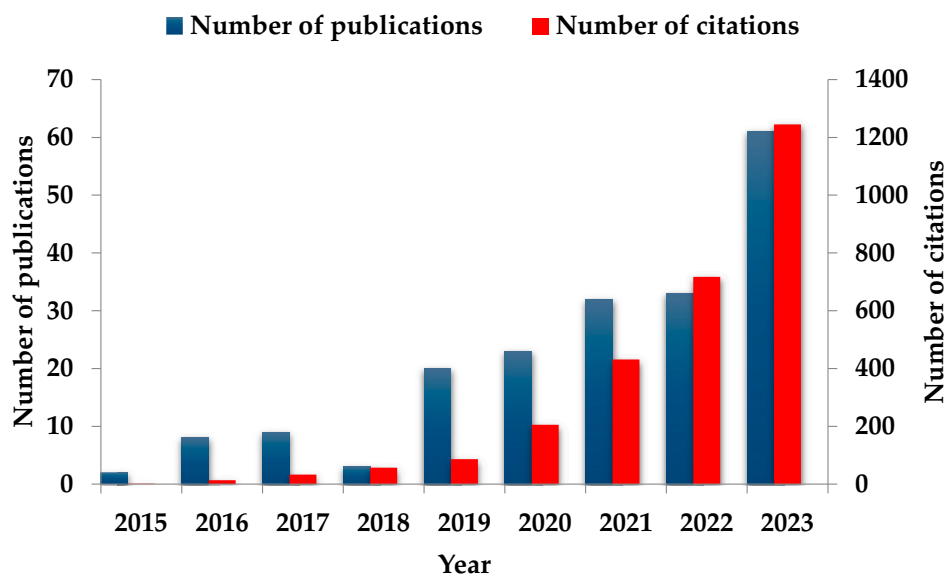


Fig. 1. Evolution of the number of publications and citations on management of food loss and food waste by Industry 4.0 technologies from 2015 to 2023 (Data were obtained from Scopus on May 6, 2024 using the following search query: Article Title, Abstract, Keywords: fourth industrial revolution OR Industry 4.0 AND food waste OR food loss).

Carrots, commonly juiced for their refreshing taste, result in a substantial number of by-products – up to 40 % of the original weight (Tlais et al., 2020). These by-products, mainly carrot pomace and peels, contain a high concentration of valuable compounds like dietary fibers (pectin), carotenoids (β -carotene), phenolic acids, minerals and vitamins. Thanks to their antioxidant, antimicrobial, anti-inflammatory and antimutagenic activities (Gulsunoglu et al., 2019), carrot by-products are gaining traction as a promising source of natural ingredients for functional foods or additives.

Date palm by-products are made of fibrous materials mainly generated during harvest (leaflets, mesh, rachis, spadix stems, and petiole) and date processing by-products (immature or spoiled dates, pollen, seeds, and press cake). Date palm biomass contains many valuable compounds, including dietary fibers, flavanols, carotenoids, phytosterols, and tocopherols (Manai et al., 2024). Many innovative extraction methods have proven to be effective in extracting phenolics, dietary fibers, flavonoids, carotenoids, and carbohydrates from date palm biomass, including ultrasonication, subcritical water extraction, subcritical fluid CO₂ extraction, subcritical fluid extraction, and microwave-assisted extraction-deep eutectic solvent extraction (Ghafoor et al., 2022; Kehili et al., 2024).

Large amounts of by-products are produced during the extraction of olive oil, including leave, pomace stones, and wastewater. Olive by-products are cheap and constitute a source of bioactive and functional compounds with a wide range of health benefits (Otero et al., 2021). These bioactive compounds include for leaves (phenolic compounds, mainly secoiridoids, flavonoids and substituted phenols); pomace (phenolic compounds, essential amino acids, sugars, organic acids, and mineral nutrients); stones (cellulose, hemicelluloses, lignin, and phenolic compounds), and wastewater (phenolic compounds, pectic

polysaccharides, and lignocellulosic residues) (Khwaldia et al., 2022).

2.2. Grain/cereal by-products

Global cereal production, surpassing 2.8 billion tons annually, generates substantial by-products (FAO, 2023). Around 400 million tons of corn, wheat, and rice harvest residues are burned yearly (Salazar-López et al., 2019). Industrial processing steps (milling, decortication, etc.) discard by-products like the pericarp, aleurone layer, and germ. Cereal by-products represent a valuable source of dietary fibers (e.g., β -glucans), proteins, bioactive peptides, lipids, phenolic compounds, phytosterols, and other health-promoting elements (Färçaş et al., 2022).

Brewer's spent grain (BSG), the main by-product of beer production, represents a significant waste stream. With global beer output reaching around 180 million tons annually, BSG production is estimated at roughly 20 % of that volume (López-Linares et al., 2020). This complex mixture, primarily composed of barley husks and seed coat fragments, boasts a valuable nutritional profile. BSG contains 30-50 % carbohydrates and 20-30 % protein, making it a potential source of valuable bioproducts (Karlsen and Skov, 2022). However, its perishable nature and complex composition have traditionally limited its use to low-value animal feed. While research explores utilizing BSG to extract valuable compounds like arabinoxylan and prebiotic oligosaccharides (Sun et al., 2024), achieving economic viability for large-scale production remains a challenge. A recent study by Sun et al. (2024) proposed a promising and cost-effective approach for industrial BSG valorization. This three-step method utilizes ammoniation pretreatment, enzymatic hydrolysis, and fermentation to convert BSG into valuable products: single-cell protein (SCP) and arabinoxyloligosaccharides (prebiotics). Notably, the process successfully yielded from every 1 kg of BSG approximately 310 g of

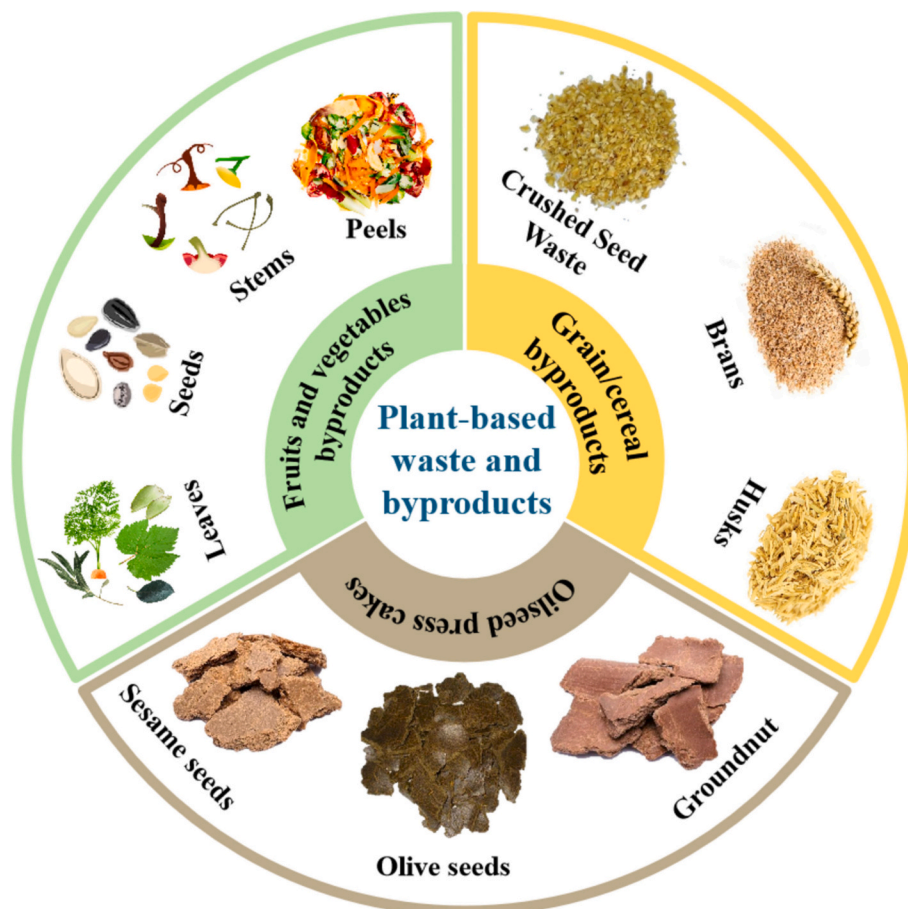


Fig. 2. Different sources of plant-based waste and by-products.

SCP and 80 g of prebiotics per kg of BSG.

Several cereal processing by-products from large-scale industries, such as BSG, rice bran, wheat bran, and corn bran, have been identified for their high protein content, making them suitable candidates for producing hydrolysates with promising biological effects (Tlais et al., 2020). Extensive research findings emphasize the feasibility of deriving bioactive peptides from proteins sourced from brewer's spent grain and rice bran, underscoring the significant value of these extracted proteins from industrial by-products. These bioactive peptides, composed of 2–50 amino acids, exert beneficial effects on human health and food products, including lowering blood pressure, scavenging free radicals, and exhibiting antimicrobial properties (Gobbetti et al., 2018).

Cereal waste and by-products serve as rich sources of essential vitamins and micronutrients, primarily concentrated in the bran and cereal germs. These processing discards are rich in both macrominerals (calcium, magnesium, etc.) and essential trace minerals (zinc, iron, etc.) (Z. Jin et al., 2022). Additionally, cereal by-products contain significant quantities of B vitamins, including niacin, pantothenic acid, and thiamine (Verni et al., 2019). Notably, BSG primarily contains choline at 1800 mg·kg⁻¹, followed by smaller amounts of niacin, pantothenic acid, riboflavin, and thiamine. Recovering valuable bioactive compounds from BSG requires careful selection of processing methods to maximize their utilization (Lech and Labus, 2022). Common techniques focus on converting BSG's cell wall components (cellulose and hemicellulose) into fermentable sugars through chemical and enzymatic hydrolysis. This process liberates functional compounds trapped within the BSG structure. Similarly, strategic extraction methods and pre-treatments can significantly enhance the yield of vitamins and minerals from cereal by-products (Lech and Labus, 2022).

Rice bran boasts the highest reported lipid content among cereal brans at nearly 19 % dry weight, followed by wheat germ and corn bran (Górnaś et al., 2016). These lipids are rich in health-promoting unsaturated fatty acids, particularly oleic and linoleic acids (Xu et al., 2021). Additionally, rice bran contains a variety of bioactive compounds like tocopherols, squalene, and phytosterols, which contribute to its antioxidant and anti-inflammatory properties (Xu et al., 2021). A by-products of wheat milling, wheat germ is a significant source of valuable nutrients, generating an estimated 25 million tons globally each year (Fărcaș et al., 2022). It boasts a wealth of health-promoting compounds, including biopeptides, polyunsaturated fatty acids, and functional components like sterols, tocopherols, and antioxidants (Tlais et al., 2020). On the other hand, wheat bran, a by-products of wheat milling (around 15 % of the grain's weight) is rich in fiber, protein, and healthy fats. Wheat bran is known to aid digestion, reduce the risk of chronic diseases, and potentially have anti-cancer effects (Deroover et al., 2020). Wheat bran's valuable composition includes dietary fibers, polyphenols, sterols, and β-glucans, contributing to its health properties (Fărcaș et al., 2022). Corn bran is rich in phenolic acids, with ferulic acid being the most abundant, followed by vanillic, caffeic, and other acids (Fărcaș et al., 2022). Additionally, corn bran is rich in anthocyanins, making them valuable food pigments (Tlais et al., 2020).

2.3. Oilseed press cakes

Beyond their primary role in oil production, oilseed processing generates valuable by-products known as oilseed meals. These meals can boast a protein content ranging from 15 to 50 % (Vinayashree and Vasu, 2021) and hold significant potential as protein sources for human nutrition. Furthermore, oilseed by-products offer economic value due to their content in bioactive compounds. Pumpkin seeds, for example, contain significant levels of carotenoids, phenolics, tocopherols, and phytosterols (Rabrenović et al., 2014). Flaxseeds are rich in essential fatty acids and lignans, while chia and sesame seeds boast polyunsaturated fatty acids, tocopherols, and lignans (Gahfoor et al., 2018). Grapeseeds, also contain essential fatty acids, tocopherols, and catechins (Al Juhaimi and Özcan, 2018). The protein content of these oilseed by-

products can reach up to 40 %, with in vitro protein digestibility ranging from 70 to 85 %. Analysis of anti-nutritional factors revealed low phytic acid content across all studied seed meals, but high tannin content was observed specifically in grapeseed meal (Sá et al., 2021).

Extracted via cold pressing, chia seed oil stands out as a nutritious botanical source rich in omega-3 fatty acids. This oil boasts not only high levels of these essential fats but also a significant protein and fiber content (Ferreira et al., 2023). Moreover, the cold-pressing process yields a cost-effective by-product, rich in polysaccharides, which further enhances the overall resource potential of chia seeds. Notably, a natural gelling agent, a gum, can be extracted from this by-product as a substitute for gelatin and whey protein in bigel production (Alkabaa et al., 2024).

Defatted sunflower flour, a by-product of extracting oil from sunflower seeds, is traditionally utilized in animal feed. Nonetheless, it offers a plentiful and economical protein source suitable for human consumption. Sunflower seeds are rich in storage proteins, with 11S globulins being the most abundant (accounting for 60–80 % of total protein), followed by 2S albumins (20–30 %). Tonolo et al. (2024) investigated bioactive peptides derived from sunflower protein isolate (SPI), obtained from defatted sunflower flour. Following simulated gastrointestinal digestion, the resulting peptide-enriched fraction exhibited antioxidant properties in zebrafish, a live animal model (Tonolo et al., 2024).

Though the exact composition can vary depending on the variety, hempseeds boast an impressive nutritional profile. They pack a powerful punch with 30–35 % of their weight in healthy fats (lipids), 20–25 % protein, 30–35 % dietary fiber, and a wealth of essential minerals (Lan et al., 2019). Rich in essential fatty acids, primarily linoleic acid and α-linolenic acid, as well as α- and γ-tocopherols, hempseeds are prized for their high oil content and unique fatty acid profile. Interestingly, the cake obtained from hempseed oil extraction serves as a valuable source of proteins, demonstrating favorable technofunctional properties such as a high solubility index, emulsifying ability, and stability (Mendoza-Pérez et al., 2024).

Due to their protein content, olive seeds, separated from the olive flesh during the pitting step, have been used for extracting protein hydrolysates rich in bioactive peptides through enzymatic processes. These hydrolysates have demonstrated antioxidant, antidiabetic and cholesterol-lowering properties. Moreover, they showed stability during simulated intestinal absorption using human intestinal cell models (Bartolomei et al., 2024). Another by-product of olive oil production, olive pomace, is the solid residue derived from the process. It retains a notable quantity of residual oil and offers a source of triterpenes. These include maslinic and oleanolic acids (triterpenic acids) as well as erythrodil and uvaol (triterpene dialcohols) (Schmidt et al., 2023).

Prickly pear seeds, comprising 2–3.8 % of the fruit, yield oil rich in unsaturated fatty acids, primarily linoleic acid (Al-Naqeb et al., 2021). The oil also boasts significant levels of vitamins (tocopherols, vitamin K1) and sterols (β-sitosterol being dominant). Notably, γ-tocopherol is the major tocopherol, outnumbering α and δ forms (Al-Naqeb et al., 2021). However, the oil's composition, especially fatty acids and tocopherols, varies depending on geographic origin (Ettalibi et al., 2021). Prickly pear seed oil exhibits antioxidant and antimicrobial properties, antidiabetic potential, lipid-lowering effects, in vitro anticancer activity, anti-inflammatory properties, anti-ulcer effects, and protection against UV radiation (Al-Naqeb et al., 2021; Ferreira et al., 2024). Prickly pear seed pomace, a by-product of seed oil extraction, is abundant in phenolic compounds and exhibits potent antioxidant properties, along with modest anti-inflammatory and anti-tyrosinase activities. Phenolic extracts obtained through an autohydrolysis process hold the potential for use in cosmetic formulations due to their effective UV-screening abilities (Ferreira et al., 2024).

Cold-pressed Sacha Inchi seed oil, extracted from the seeds of the oleaginous Sacha Inchi plant, is prized for its richness in essential fatty acids like α-linolenic acid and tocopherols. This oil has gained

recognition as a potential dietary aid for cardiovascular health, including coronary heart disease, hypercholesterolemia, and hypertension (Mhd Rodzi and Lee, 2022).

Despite being a by-product of Sacha Inchi oil production, the seed husk contains various phenolic compounds, such as protocatechuic acid, hydroxycinnamic acid derivatives, and ferulic acid, showcasing anti-oxidant and anti-aging properties (Poomanee et al., 2024).

3. Emerging techniques for recovering bioactive compounds

Emerging extraction methods aligned with the “Green Extraction Process” concept have been developed, prioritizing renewable resources, alternative solvents, energy efficiency, waste minimization, process safety, and high-quality extract production (Chemat et al., 2012). The use of green extraction methods such as microwave assisted extraction (MAE), ultrasound-assisted extraction (UAE), enzyme-assisted extraction (EAE), supercritical fluid extraction (SFE), pressurized liquid extraction (PLE), pulsed electric field (PEF), and membrane separation technology (MST), to convert plant-based waste and by-products into valuable bioactive compounds can promote a more sustainable model by reducing waste and generating new revenue streams.

The MAE technique utilizes non-ionizing microwave radiation (frequency 0.3–300 GHz) to rapidly heat solvents through interaction with polar molecules within plant matrices, facilitating analyte extraction (Bitwell et al., 2023). Essential oils from citrus peels, phenolic compounds from apple pomace, grape skins, olive mill wastewater, cereal brans, soybean meal, rapeseed meal, and sunflower seed meal, pigments from tomato and carrot peels, and blueberry pomace, and polysaccharides from apple pomace, citrus peels, sugar beet pulp, wheat and oat brans, corncobs, and rice husk are examples of successfully extracted bioactive compounds by MAE (Alvi et al., 2022). MAE offers significant advantages, including reduced processing time, solvent consumption, and energy usage, while producing higher-quality extracts at lower costs.

UAE employs high-frequency sound waves to disrupt plant matrices and enhance the extraction of target compounds. Cavitation, induced by the rapid formation and collapse of bubbles, generates localized high temperatures and pressures, leading to cell disruption and increased mass transfer (Kumar et al., 2021). UAE is an efficient technique with reduced extraction time and energy consumption, particularly suited for extracting thermolabile compounds, though it may induce degradation due to the formation of free radicals (Hasan et al., 2024). UAE has been used to extract polysaccharides from wheat bran, rice husk, apple pomace, pomegranate and grape skins; anthocyanins from blueberry and raspberry pomace, cherry pomace, and blackcurrant pomace; and carotenoids from tomato peels, papaya peel and pulp, mango pomace, pomegranate waste, and carrot pomace (Martins et al., 2023).

EAE utilizes enzymes to disrupt plant cell structures, enhancing the release of phytochemicals. This green and efficient method reduces the use of solvents and energy compared to traditional extraction techniques. EAE has been employed to improve the extraction of a wide range of compounds, including phenolic compounds such as flavonoids and anthocyanidins from saffron tepals, wheat bran, pistachio green hulls, bean shells, and brewers' spent grain; carotenoids from tomato waste; and biopolymers from grapefruit peel, buckwheat husk, palm kernel, and soybean meal (Streimikyte et al., 2022). However, challenges include high costs, scalability issues, and enzyme limitations in breaking down plant cell walls.

SFE is a sustainable and eco-friendly method for extracting valuable compounds from plant-based waste and by-products. This process utilizes solvents, primarily carbon dioxide (CO₂), under specific temperature and pressure conditions that transform the CO₂ into a supercritical fluid. This state allows for efficient solubilization of lipophilic compounds (e.g., carotenoids, essential oils, and flavonoids) due to the fluid's unique properties, including low density, viscosity, and surface tension, combined with high diffusivity (Gan and Baroutian, 2022).

Despite its efficacy in extracting heat-sensitive compounds, SFE requires precise control of operating parameters (e.g. solvent flow, pressure and temperature) and faces challenges related to cost and continuous operation.

PLE employs solvent extraction at high temperatures and pressures to enhance analyte recovery while maintaining the solvent in a liquid state throughout the process. This technique disrupts cellular structures, liberating both polar and non-polar compounds, including essential oils from coriander seeds and citrus peels, carotenoids from tomato pomace and carrot roots, anthocyanins from grape skins and blackberry fruits, phenolics from oregano by-products and olive leaves, proteins from soybean meal, and carbohydrates from potato peels (Barp et al., 2023). PLE is a rapid, efficient, and solvent-saving technique that enhances analyte extraction through increased temperature and pressure. However, the high cost of specialized equipment limits its widespread application.

By applying short, high-voltage pulses, PEF induces electroporation, creating pores in cell walls and enhancing mass transfer. This non-thermal, green, and cost-effective process has been effectively used to release polyphenols, flavonoids, and anthocyanins from plum stems and pomace, white mulberry leaves, olive leaves, blueberry press cake, apricot pomace and kernels, quince peels, chicory roots, cocoa bean shells, custard apple leaves, and flaxseed hulls; steroidal alkaloids from potato peels; polysaccharides from jackfruit waste and corn silk; and proteins from rapeseed stems and leaves (Chatzimitakos et al., 2023).

Membrane filtration, including nanofiltration, microfiltration, and ultrafiltration, offers a sustainable and solvent-free approach for isolating phytochemicals from plant-based waste and by-products. MST has gained significant attention in the industry for its ability to concentrate, purify, and separate valuable compounds. Offering high recovery rates and selectivity, MST is particularly effective for recovering phenolics (e.g., phenolic acids, catechol, tyrosol, and hydroxytyrosol) from wine lees, grape marc, fruit seeds, and corn, olive, and artichoke wastewaters (Castro-Muñoz et al., 2020).

Industry 4.0 technologies offer significant opportunities to enhance the sustainability of green and novel extraction processes. By integrating digital tools for energy efficiency, waste reduction, process optimization, and traceability, these processes can contribute to a more circular and environmentally friendly manufacturing landscape. Real-time monitoring, predictive maintenance, renewable energy integration, data-driven decision-making, automation, and blockchain technology are key elements in achieving these goals.

4. Plant-based waste and by-products and food sustainability

Food sustainability and more broadly sustainable development requires an integrated approach that takes into consideration environmental and social concerns along with economic development. Every year, a staggering 1.3 billion tonnes of food are wasted or lost worldwide (Durán-Sandoval et al., 2023). According to Roy et al. (2023) developed countries, with a total population of 1.4 billion, threw away around 670 million tonnes of edible food, while developing countries, with a population of 6.2 billion, contributed around 630 million tonnes. These figures stand in stark contrast to the global reality, where between 690 and 828 million people suffer from hunger, and 3 billion cannot afford a healthy diet (FAO et al., 2022; Roy et al., 2023).

4.1. Environmental and social dimension

Reducing food waste not only benefits the environment but also contributes to various social agendas by reducing greenhouse gas emissions, enhancing ecosystem services, and enforcing sustainable resource management policies (Table 1). Food waste accounts for 8–10 % of global greenhouse gas emissions and ranks as the third-largest emitter globally, following China and the United States (S. You et al., 2022).

Table 1

Social, environmental and economic dimensions of waste and by-products valorisation.

Valorisation of plant waste	Social impact	Environmental impact	Economic impact
Plant waste-based compost	- Community involvement in waste management hence more sustainable environmental and agricultural practices - Educational opportunities on waste reduction and organic farming	- Reduced landfill waste and greenhouse gas emissions - Lower use of landfill sites and environmental risks associated with the accumulation of waste hence low pollution - Improved soil fertility.	Low fertilizer costs
Extract of bioactive compounds from plant waste	Availability of healthier and sustainable foods containing bioactive compound extract	- Reduced environmental impact due to the decrease in synthetic additive application - Reduced environmental pollution by minimizing landfill contributions and promoting a circular economy. - Higher efficiency of bioactive recovery and lower ecological footprint using eco-friendly extraction methods	- Additional revenue streams for industries by leading to the development of new, cost-effective food products. - Extended food shelf life by incorporating natural bioactive compounds as food additives.
Plant waste as animal feed	- Better food security by redirecting surplus food to livestock, thus enhancing food availability for human consumption - More available jobs in waste processing and feed production	- Improved nutritive value and digestibility of low-quality raw materials using bioprocess techniques - Reduced environmental impact of livestock production	More economically viable livestock production due to the reduced feed costs
Plant waste-based bioplastics	Safer alternative for food packaging	- Reduced plastic pollution - More eco-friendly due to their biodegradability - Lower Carbon Footprint	Alternative materials and energy sources hence reducing overall costs
Plant waste-based bioenergy	- Less competition with food crops - Higher worker safety and community health by applying sustainable practices in bioenergy production	- Production of renewable energy - Possibility of recycling nutrients and reduction in the waste volume - Reduced dependency on fossil fuels and carbon emissions, thereby promoting the development of a low-carbon economy	- Increased biogas production and improved profitability of AD systems by reducing the costs associated with waste management - Cost-Effective Resource Utilization - Job Creation

Utilizing industrial food waste for higher value-added purposes could save significant CO₂ equivalent emissions (RedCorn et al., 2018; Roy et al., 2023). However, caution is advised regarding energy recovery from food waste incineration due to its high moisture content (Kim and Song, 2014). Anaerobic digestion is identified as a preferable option over incineration, with decentralized anaerobic digestion offering the highest environmental benefit. Recycling and reusing agricultural waste and animal dung for biogas production are promising avenues for sustainable waste management and energy production (Koul et al., 2022).

The environmental impact of food waste management varies depending on the method employed, with landfilling having the highest impact and animal feeding the least. Discrepancies among studies are attributed to methodological choices and geographical variations (Roy et al., 2023).

4.2. Economic dimension

Selecting the right approach for valorizing food waste is crucial for cost-effective management and resource conservation (Roy et al., 2023). The economic impact of food waste is closely tied to the valorization processes used, with economic benefits depending on the derived end-products (Table 1) (Kotykova and Babych, 2019). The significant economic cost of lost and wasted food, estimated at U.S. \$680 billion in developed countries and U.S. \$310 billion in developing countries, underscores the urgency of effective management strategies (Laso et al., 2020).

While some studies question the economic viability of food waste anaerobic digestion, emphasizing the need for additional revenue streams such as using biogas for vehicles or on-grid electricity to enhance economic benefits (Flotats, 2019) others assert its feasibility, particularly when supported by government subsidies (Papirio et al., 2020; Roy et al., 2023). In the United States, converting industrial food waste into lactic acids or biogas is estimated to generate significant annual revenue (Papirio et al., 2020).

Incineration is identified as having the least life cycle cost compared to other waste management options, though decentralized composting of food waste is economically beneficial, with examples like on-site incineration at Hong Kong International Airport showing promising sustainability and economic savings (Lam et al., 2018). The economic viability of food waste biorefineries hinges on factors like plant capacity, product diversity, and integration with existing infrastructure (Roy et al., 2023).

Studies have examined various economic indicators to assess the profitability of food waste recovery scenarios, highlighting the influence of market prices and economies of scale (Cristóbal et al., 2018). Integrating food waste biorefineries into existing facilities can significantly reduce production costs and enhance economic sustainability (Lohrasbi et al., 2010).

Overall, a comprehensive economic evaluation of food waste valorization requires considering waste stream availability and proximity to biorefinery locations, factors often overlooked in previous studies (Caldeira et al., 2020). Such considerations are essential for ensuring the economic sustainability of food waste management practices.

5. Potential of Industry 4.0 technologies to reduce and valorize plant-based waste and by-products

Digital technologies in the food sector is currently transforming industry operating methods (Abbate et al., 2023; Dalenogare et al., 2018). Despite the drawbacks that can be related to the implementation of these technologies (Table 2), they promise to revolutionize the management of plant-based waste food and by-products, leading to reduced input consumption, reuse, recovery, recycling and reduced waste and emissions, contributing to a more sustainable and circular food system (Fig. 3) (L. Brennan et al., 2015; Cheah et al., 2022; Hassoun et al., 2023a). These innovative technologies can potentially provide the

Table 2

Main benefits and drawbacks of food industry 4.0 technologies in plant waste and by-products management.

Industry 4.0 technology	Benefits	Drawbacks
Smart sensors	• Provide continuous data on critical parameters such as temperature, moisture, and spoilage, enabling timely interventions to prevent waste. • Improve the accuracy of data collection, leading to better decision-making in processing and storage of plant-based materials. • Help in optimizing the use of resources by ensuring conditions are ideal for minimizing waste and enhancing by-product quality.	• Installation and calibration of smart sensors can be expensive, especially for large-scale operations. • The volume of data generated can be overwhelming, requiring sophisticated systems to analyze and act on the information. • Some sensors require regular maintenance and calibration to ensure accuracy, which can be labor-intensive and costly.
IoT	• Enables seamless communication between devices, ensuring all parts of the supply chain are synchronized to reduce waste. • IoT devices collect vast amounts of data, which can be analyzed to optimize processes and reduce inefficiencies. • Allows for real-time monitoring and adjustments from remote locations, reducing loss and waste.	• Increased connectivity can lead to vulnerabilities in the system, making it susceptible to cyber-attacks. • Integrating IoT devices with existing systems can be challenging, especially if there is a lack of standardization. • The initial setup of IoT infrastructure can be costly, particularly for small to medium-sized enterprises.
3D printing	• Allows for the creation of custom-designed products from plant-based by-products, opening new markets and opportunities. • Can use plant-based waste materials as raw materials for printing, effectively reducing waste. • Enables rapid prototyping and testing of new products made from plant-based by-products.	• Not all plant-based by-products are suitable for 3D printing, limiting the scope of its application. • The cost of 3D printers and the materials needed can be high, making it a less viable option for some businesses. • Requires specialized knowledge to design and print products effectively, which can be a barrier to adoption.
AI	• Can analyze historical data to predict and prevent waste, optimizing processes before issues arise. • Automates complex tasks such as sorting and processing, reducing human error and increasing efficiency in managing plant-based by-products. • Provides insights and recommendations that improve decision-making in waste management and by-product utilization.	• Could require large amounts of data to function effectively, and poor-quality data can lead to inaccurate outcomes. • Could raise ethical questions about data privacy and the potential displacement of human jobs in the industry. • Implementing and maintaining AI systems could require specialized technical expertise, which can be difficult to source.
Robotics	• Can work faster and more accurately than humans, especially in repetitive tasks such as sorting and packaging plant-based by-products. • Ensure uniformity in processing, leading to higher quality by-products and reduced waste. • Automating labor-intensive tasks can lead to significant cost savings over time.	• The cost of purchasing, installing, and programming robots can be prohibitively expensive for many food businesses. • Operating and maintaining robotic systems requires advanced technical knowledge and skills. • The automation of tasks previously performed by humans could lead to job losses, which may have

Table 2 (continued)

Industry 4.0 technology	Benefits	Drawbacks
Blockchain	• Provides a secure, tamper-proof record of transactions and processes, ensuring full transparency in the supply chain. • Blockchain's decentralized nature makes it highly secure, protecting sensitive data from unauthorized access or tampering. • Streamlines record-keeping processes, reducing the likelihood of errors and fraud.	• broader social and economic implications. • As more transactions are added, blockchain networks can become slower and less efficient. • Blockchain technologies, particularly those using proof-of-work consensus mechanisms, can be energy-intensive. • Implementing blockchain solutions can be complex and costly, especially for smaller food businesses without substantial resources.
Digital twins	• Allow for the simulation of processes in a virtual environment, identifying inefficiencies and optimizing waste management strategies. • Can predict equipment failures and maintenance needs, reducing downtime and preventing waste. • Provides real-time data and analytics, enabling dynamic adjustments to processes to reduce waste.	• Setting up digital twins could require significant investment in both technology and expertise. • Their effectiveness depends on the accuracy of the input data, as inaccurate data can lead to incorrect simulations and poor decision-making. • The technology and infrastructure required for digital twins can be expensive, making it less accessible for smaller food companies.

achievement of increasingly high levels of efficient industrial production and the optimization of processes throughout the entire production supply chain (Müller et al., 2018). However, creating and incorporating new technologies to tackle food waste issues alone is not enough. Reduction in waste generation deserves significant emphasis. The automation of certain processes and the combination of data with IoT supports remote monitoring and control which eliminates the risks and hazards associated with solid waste labor workers. With big data and AI, advanced analytics can be synthesized to provide a deeper understanding and reveal underlying relations within them. Technologies like machine learning (ML) can be used to train algorithms in image recognition to automate some of the waste segregation processes (Raju and Thiagarajan, 2020). Large scale manufacturers can use data harnessed through IoT to gain a clearer picture on how their products are used and track disposal patterns to nurture future designs for a truly circular economy.

This part of the review aims to present some relevant applications of the Industry 4.0 technologies, such additive manufacturing (3D printing), AI, IoT, and smart sensors, in the management of loss and waste valorization (Table 3) (Hassoun et al., 2023c; Hassoun et al., 2022c), focusing on the plant-based supply chain.

5.1. Artificial Intelligence (AI)

Neethirajan (2023) defines AI as systems that precisely understand data, learn from it, and apply it to carry out particular and customized activities that mimic human behavior. In the contemporary era of Food Industry 4.0, AI can be used for a lot of tasks in the food industry, in instance, to track and improve food production and supply chains, promote circular economy programs, among others. Therefore, AI can lead to a more healthy and ethical food system by its potential to optimize the use of resources and reduce greenhouse gas emissions (Sharma et al., 2021) associated to plant-based industry. Recent advancements in AI technologies such as deep learning, image recognition, ML, and

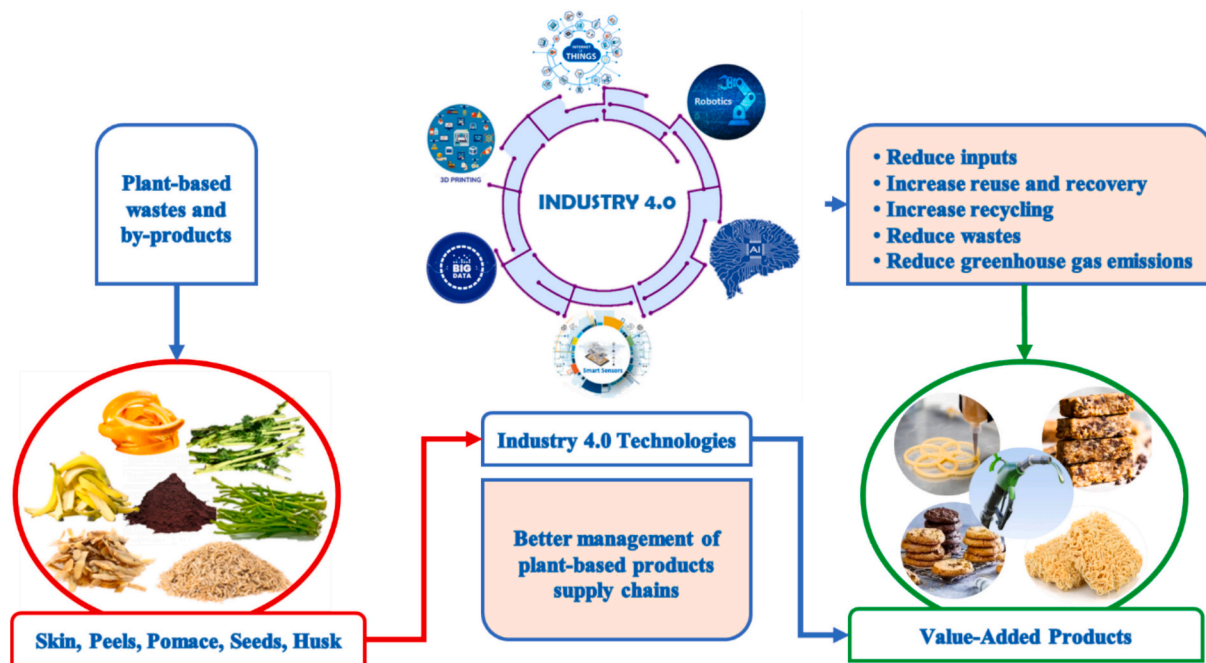


Fig. 3. Industry 4.0 technologies involvement in plant-based waste and by-products management.

natural language processing offer promising prospects for reshaping the global food system (Davenport, 2018).

For instance, by utilizing meteorological data, crop yields, and consumer demands, AI enhances agricultural practices and food supply chain management, resulting in higher yields and less waste along the supply chain (P. Kumar et al., 2022).

Waste sorting is tremendously important because it is an effective way to achieve waste reduction. ML and computer vision are suitable techniques to enable the development of waste material recognition and classification systems. These systems can be trained to distinguish various types of waste, including plastic, glass, paper, and even organic waste from fruit and vegetable. In the study conducted by Zhang et al. (2021), the authors demonstrated the effectiveness of a deep learning algorithm for automatic classification of food waste based on images captured by surveillance cameras in restaurants. This approach facilitates the identification and quantification of generated waste types, thereby easing their management and subsequent processing. Additionally, Ortega-Fernández et al. (2020) examined the use of deep learning techniques for intelligent waste management, focusing on accurate waste classification and optimization of sorting processes.

Regarding biogas production, Yang et al. (2021) utilized ML models employing the Adaptive Neuro-Fuzzy Inference System (ANFIS) and Least Squares Support Vector Machine (LSSVM) to forecast biogas generation from diverse organic waste origins, including vegetables, fruits, and food waste. These models identified key factors influencing biogas production and optimized anaerobic fermentation conditions, thus contributing to better utilization of organic resources and reduction of greenhouse gas emissions. Similarly, Xiao et al. (2021) explored the use of state retention neural networks (NARX) and backpropagation (BP) methods to predict biogas production during the conversion of organic waste into compost. By utilizing data on environmental conditions, waste characteristics, and microbial activities, their model accurately estimated composting yield and identified optimal conditions to maximize biogas production.

AI can be utilized to optimize waste valorization processes such as composting or methanization. For instance, predictive models based on AI can be developed to monitor and control composting conditions to optimize the organic waste decomposition process and produce high-quality compost. A study conducted by Chiu et al. (2022) employed

ML techniques (based on random forest and long short-term memory) to predict biogas production from food waste based on factors such as waste composition, temperature, and humidity. These predictions enable the adjustment of methanization process parameters to maximize energy yield.

Another significant application is presented by You et al. (2017), who compared several ML models for online classification of the calorific value of municipal solid waste burned in circulating fluidized bed incinerators. By utilizing techniques such as artificial neural networks (ANN), support vector machines (SVM), and random forests (RF), they developed an online classification system capable of rapidly identifying the composition and energy value of burned waste, thereby optimizing the combustion process and energy production. Furthermore, Neto et al. (2021) used ANN for modelling production biogas from fruit and vegetable waste and predicting biogas yield and composition. The response surfaces of the ANN model were found to be useful for defining the optimum region in biogas production. It was observed that bio-digestion of fruit and vegetable waste could lead to a highest cumulative biogas production.

In summary, AI provides innovative solutions for optimizing waste valorization processes, particularly through material recognition, classification, and biogas production forecasting. These advancements contribute to better management of organic resources and the reduction of greenhouse gas emissions, aligning industrial practices with the goals of a circular and sustainable economy.

5.2. The Internet of Things (IoT)

IoT systems provide a wide range of applications in the manufacturing industry to handle supply chain inventory and data, operate production equipment, and optimize energy consumption. IoT enables objects to be detected and controlled remotely via the available network infrastructure, providing opportunities to integrate the physical world more directly into information systems, thus improving efficiency and precision. It describes the connection of various devices and physical objects all over the world via the Internet (Branger and Pang, 2015; Gokhale et al., 2018). While various structural layers have been delineated for IoT based on specific application domains, the predominant focus in most studies centers on establishing three primary layers. These

Table 3

Industry 4.0 technologies applied to reduce and valorize plant-based waste and by-products.

Industry 4.0 technologies	Techniques used	Plant-based waste and by-products	Main issue addressed	Value-added products	References
Additive manufacturing or 3D Printing	3D printing with extrusion-based technology	Grape pomace and broken wheat	Improved utilization of food industry waste/by-products through the production of customized, nutritionally enhanced foods	Functional cookies with increased protein, dietary fiber, and antioxidant properties	Jagadiswaran et al. (2021)
	Printing process	White Grape pomace	Enhancing the nutritional and antioxidant properties of biscuits while evaluating rheological and sensory changes	Biscuits with increased dietary fiber, phenolic content, and antioxidant properties	Mildner-Szkudlarczyk et al. (2013)
	3D printing using Direct Ink Writing (DIW)	Orange peel	Upcycling orange peel waste into high-value, nutritionally enriched food products	Edible 3D-printed snacks with antioxidant properties and bioflavonoids	Leo et al. (2022),
	3D printing of gelatin gels with orange by-products (OBP)	OBP	Valorization of agri-food industry waste and reduction of food waste	Fiber-enriched 3D-printed snacks with enhanced bioactive compounds and improved nutritional value	Molina-Montero et al. (2023)
	3D printing of nutrient-rich noodles	Potato peel waste from potato processing industries	Effective utilization of food waste and promotion of sustainable food production	Nutrient-rich, 3D-printed instant noodles with high consumer acceptance	Muthurajan et al. (2021)
	3D printing of high-fiber snacks	Potato processing by-products	Utilization of potato by-products to enhance the fiber content and quality of snacks	High-fiber, 3D-printed yam snacks with optimized physical properties for air frying	Feng et al. (2020)
	3D food printing	Spinach stems and kale stems	Valorization of vegetable waste into edible, consumer-accepted products	Edible food products with familiar taste, texture, and appearance derived from vegetable waste	Pant et al. (2023)
	3D food printing with guar gum (GG)	Banana peel and sugarcane bagasse	Conversion of non-printable agricultural waste into printable materials for food packaging	Customized 3D-printed food packages made from banana peel and sugarcane bagasse	Nida et al., 2022
Smart sensors	3D Food Printing	Jackfruit seeds	Overcoming barriers to consumer acceptance of food waste as edible products	Customized food items (e.g., snacks) made from jackfruit seeds	Wong et al. (2022)
	Scanning Electron Microscopy, Energy Dispersive Spectra, Fourier Transform Infra-Red, and X-Ray Diffraction	Orange peel fibers	Improving the acoustic properties of waste materials for noise reduction	Graded acoustic materials for noise absorption	Nath and Singh (2021)
Digital twins	RFID	food waste	Monitoring and reducing individual food waste by tracking consumption and providing rewards	Reduced food waste, increased awareness, and incentivized responsible consumption	Gayathri et al. (2021)
		Medicinal and aromatic plants	Increasing sustainability and utilization of plant components usually discarded	Valuable metabolites, sustainable production strategies.	Kümmritz et al. (2024)
Artificial Intelligence	ML, ANFIS and LSSVM	Vegetable Food and fruits	Modelling and predicting biogas production from various inputs	Accurate biogas production	Yang et al. (2021)
	NARX and BP Hybrid Neural Networks	Organic waste	Accurate estimation of CH ₄ yield via Direct Electron Transfer (DET) in microbial electrolysis cells (MECs), improving the understanding of methanogenesis mechanisms	Enhanced CH ₄ yield prediction, better systematization of methanogenesis processes	Xiao et al. (2021)
	Hybrid Modelling (Random Forest + LSTM), Key Factor Analysis	Municipal food waste	Improving the prediction accuracy of biogas production by identifying key input factors and optimizing the process using hybrid machine learning models	Improved biogas output prediction, enhanced process control, reduced costs	Chiu et al. (2022)
	SVM, ANFIS, MLP, and RF	Municipal Solid Waste (MSW)	Predicting the heating values (HVs) of burning MSW in circulating fluidized bed (CFB) incinerators	Accurate prediction models for HVs in CFB incinerators	You et al. (2017)
	ANN	Food waste, fruit and vegetable waste, and their blends	Predicting cumulative biogas production under varying conditions (e.g., OLR, temperature)	Optimized biogas production models for different organic wastes	Neto et al. (2021)
Internet of Things	IoT-based real-time monitoring system using load cells and image processing with CNN	Potato waste in the potato processing industry	Automated and accurate tracking of potato waste and its causes	Improved waste management, cost savings, enhanced quality control	Jagtap et al. (2019)
	IoT-based system with RFID sensors and cloud computing	Food waste	Monitoring and reducing individual food waste by tracking consumption and providing rewards	Reduced food waste, increased awareness, and incentivized responsible consumption	Gayathri et al. (2021)

layers consist of: i) the device layer encompassing sensors, RFID, and other physical devices responsible for data collection, ii) the network layer comprising various network communication protocols utilized for transmitting data collected by the device layer, and iii) the application layer, which includes IoT applications and services (Bouzembrak et al.,

2019; Friha et al., 2021; Hassoun et al., 2022c).

The development of IoT intersects with the digitalization of the food processing and production sectors, offering insights into methodologies to reduce waste and optimize industry processing (C. S. Brennan, 2024).

In the context of transforming plant-based waste, the main

application of IoT lies in controlling and measuring bioprocess parameters. Indeed, precise control of the numerous components influencing bioprocesses is crucial to meet the growing demands of research and production. These components include the concentration of nutrients and metabolites, cell density, pH, oxygen level, humidity, and temperature (Neeba et al., 2021). Real-time measurement of these parameters is complex but essential. Spectroscopic methods such as UV-visible, mid-infrared, near-infrared, fluorescence, and Raman show potential for real-time monitoring of various parameters but suffer from limitations such as poor discernment and data processing complexity (Rusli et al., 2021). The IoT emerges as a promising solution, enabling real-time monitoring of key parameters such as humidity, temperature, pH, and oxygen level, thanks to innovative sensors and microfabrication techniques (Chau et al., 2021; Hassoun et al., 2022c; Van et al., 2022). Recent studies have shown that IoT facilitates the monitoring and management of bioprocesses, particularly in food waste valorization (Kalsoom et al., 2021; Kazemi et al., 2020; Nath and Singh, 2021). However, challenges remain, such as the complexity of sensor and data network architecture, as well as potential security and compatibility issues (Melikoglu, 2020). Only a few studies have described the application of IoT for controlling plant-based waste in food processing process. For instance, Jagtap et al. (2019) developed an automated, real-time system based IoT to monitor the global quantity of waste and the reasons for its production in real time throughout the potato transformation and processing industry, employing advanced image processing and sensor technology. The authors utilized the images captured and processed by a specially positioned camera to recognize damaged and unusable products, while a digital scale was applied to measure their weight. Using IoT resulted in cost savings and reduced the risk of unwanted potatoes finding their way into the final food product (Jagtap et al., 2019).

In the same context, at a different level of the food production and distribution process, Gayathri et al. (2021) and Luthra et al. (2018) have also explored the use of IoT to mitigate food waste. Gayathri et al. (2021) employed RFID sensors to manage individual food waste according to a suggested model, while Luthra et al. (2018) outlined the implementation of IoT-based technologies in supply chain management within developing nations. These studies highlight the diverse applications of IoT in addressing food waste challenges across various stages of the food production and distribution process.

In conclusion, the IoT plays a crucial role in real-time monitoring and control of bioprocess parameters, which is essential for enhancing the valorization of plant-based food waste. IoT solutions not only optimize industrial processes but also reduce costs and risks, thereby strengthening the efficiency and sustainability of the food industry.

5.3. Additive manufacturing or 3D Printing

With its potential for a wide range of uses in the food business, additive manufacturing, often known as 3D printing, is an emerging technology that presents a fresh approach to food processing and the creation of novel formulas (Lv et al., 2023; Piyush et al., 2020). The method is designed to swiftly and simply create aesthetically pleasing and nutrient-dense food items with little to no human involvement and without costly tooling, resulting in more effective resource usage and little to no waste (Lee et al., 2020). A great deal of work is being done on 3D printing technologies and given the immense promise and potential applications of this technology, it is even becoming a new industrial revolution (Campbell et al., 2011).

Through the use of 3D printing technology, plant waste and by-products can be transformed in new, value-added products (Godoi et al., 2016). This approach provides the food industry with the opportunity to create innovative products tailored to consumer preferences and market demand (Lipton et al., 2015). For example, Grape pomace was collected from industries and utilized as a key ingredient of a 3D printing material for functional cookies (Jagadiswaran et al., 2021).

They noted an increase in viscosity as a function of grape content, leading to the use of a lower print speed and extrusion rate to improve printability. Similar observations were reported by Mildner-Szkudlarz et al. (2013), who optimized the printing process of dough consisting of cracked wheat flour and various concentrations of grape pomace powder. Functional cookies exhibited improved palatability, shape stability, higher levels of antioxidant properties and less anthocyanin content loss and increased fiber and protein contents.

In another study, Leo et al. (2022) designed 3D-printed snacks enriched in vitamins and nutrients by using waste orange peel. The developed snacks presented a high content in bioflavonoids and antioxidant activity. Molina-Montero et al. (2023) by using orange peel ink found that orange by-products had a significant effect on the print process and final snack quality. All the developed formulations showed a more pronounced elastic behavior and greater crunch.

Muthurajan et al. (2021) designed ready-to-cook noodles from waste potato peelings, comparing the texture, cooking quality, and consumer acceptance with commercial noodles. The resulted product proved to be rich in nutrients and suitable for various forms or layering. Moreover, potato waste has been used by Feng et al. (2020) to develop a yam-based snack using 3D printing. Additionally, the study of Derossi et al. (2018) deployed 3D printing techniques in order to create a fruit-based snack for children.

Regarding ink formulation, research conducted by Pant et al. (2023) investigated recipes for food inks formulated from green leafy vegetables such as spinach and kale stems. They characterized these wastes on their rheological properties to develop soft, easy-to-chew 3D-printed foods for patients suffering from dysphagia (difficulty swallowing). Furthermore, other researchers investigated the valorization of banana peel and sugarcane bagasse into food packaging using 3D printing (Nida et al., 2022). The paper explored the impact of combining banana peel and sugarcane bagasse in different ratios (1:1, 1:9, 9:1) along with the incorporation of guar gum for 3D food packaging personalization.

Jackfruit seeds are known to be a rich source of starch, protein, fiber, and magnesium, known for promoting good bone health and lowering blood pressure. The study of Wong et al. (2022) indicates the feasibility of 3D printing jackfruit seeds after pre-treatment with a food-grade ink suitable for 3D printing. The researchers concluded that food waste like jackfruit seeds can be designed and packaged into readily consumable snacks and meals: biscuits, cookies, wafers, and chips. 3D printing is a useful technique for transforming unconventional food components such as jackfruit seeds through aesthetic and visually appetizing shapes, thus promoting consumer acceptance.

3D printing offers promising prospects for transforming plant waste into high-value products, such as functional snacks and customized food packaging. This innovative technology not only reduces waste but also meets the growing market demand for more sustainable and personalized products.

5.4. Smart sensors

Kerry (2012) identified sensors as a device utilized to detect, locate and/or quantify energy or matter, providing a signal either for the identification or measurement of a physical or chemical property. Cloud computing services provide possibilities for data processing, storage, and collecting that let sensors reach their full potential. The integration of smart sensors with IoT, AI, and other Industry 4.0 technologies has made smart sensors a crucial component of contemporary agriculture and food supply chains, offering valuable data for executive decision-making. Smart sensors as part of a wide range of production equipment are used more and more in the food industry to monitor, control and optimize numerous manufacturing processes in real time, while at the same time improving food traceability and quality (Watson et al., 2021). As food products evolve and production and processing conditions change, smart sensors are designed to be tiny, flexible, and versatile.

Spectral fingerprinting methods, such as near-infrared spectroscopy and hyperspectral imaging, and optical fibers form the foundation of the majority of sensors used in the food industry. Among the various uses for such non-destructive optical sensing technologies are quality and process control, food shelf-life and chemical composition prediction, and parameter monitoring throughout processing (Hassoun et al., 2022a).

The unique structural patterns inside fibrous materials such as orange peel have attracted the interest of Nath and Singh (2021) to develop a particular concern to design some building acoustic materials and numerous other technological products using sensors. This study examined untreated and treated orange peel fibers of different particle sizes by Scanning Electron Microscopy, Energy Dispersive Spectra, Fourier Transform Infra-Red, and X-Ray Diffraction indicating that ultrasonic deformation of cellulose and anticellulose makes composites a suitable acoustic material. Moreover, Gayathri et al. (2021) employed RFID sensors to manage individual food waste according to a suggested model. They studied the integration of sensors in a waste collection system with digital tracking of measurements, including the weighing of bags and containers before transfer, it is possible to create a micro-analysis mechanism to accurately report the amount of waste exported and disposed of to recover food waste.

In closing, 3D printing offers promising prospects for transforming plant waste into high-value products, such as functional snacks and customized food packaging. This innovative technology not only reduces waste but also meets the growing market demand for more sustainable and personalized products.

5.5. Digital twins

Digital twins involve creating digital representations of real-world products, processes, or physical objects. These representations integrate various technological advancements like IoT and AI to synchronize physical activities with the virtual world. Digital twins come in statistical, data-driven, and physics-based models (Defraeye et al., 2021; Tao et al., 2019; Van Der Burg et al., 2021; Verboven et al., 2020). They hold the potential to enhance knowledge and streamline decision-making across different sectors such as agriculture and food processing (Defraeye et al., 2021; Van Der Burg et al., 2021; Verboven et al., 2020). Furthermore, they could aid in predicting postharvest changes in food quality and optimizing supply chains to minimize food loss (Defraeye et al., 2021; Onwude et al., 2024).

Only a few studies have described the application of digital twins in Transforming plant-based waste into valuable products. For instance, Kümmritz et al. (2024) investigated the potential of digital twins to enhance sustainability in the cultivation and utilization of medicinal and aromatic plants. Their research highlights how these precise virtual models can not only optimize agricultural practices for these plants but also open new avenues in the extraction methods of active compounds. Digital twins provide detailed simulations and analyses of plant growth conditions, enabling farmers to optimize resource use (such as water and nutrients) reduce waste, and increase yields. Additionally, these virtual models offer innovative solutions for more efficient and environmentally friendly extraction processes of beneficial plant compounds, such as essential oils and medicinal substances. Moreover, digital twins present a valuable alternative for recycling organic waste generated by these plants by converting it into biofertilizers. This approach not only minimizes waste but also enriches the soil, thereby closing the production cycle in a sustainable manner. Overall, digital twins hold significant promise for making the cultivation and use of medicinal and aromatic plants more sustainable and efficient while providing novel solutions for organic waste management.

Finally, digital twins represent an emerging technology with considerable potential to optimize the cultivation and valorization of medicinal and aromatic plants. By simulating growth conditions and improving extraction processes, these virtual models not only reduce waste but also enrich the soil, fostering more sustainable agriculture and

a more environmentally friendly production cycle.

5.6. Robotics

In recent years, major food firms have increasingly integrated robotics into their food manufacturing plants, significantly contributing to the reduction and valorization of food loss and waste, particularly of plant origin (Duong et al., 2020). In the agricultural and food industries, robots now perform a wide range of tasks, especially when combined with intelligent sensors, such as advanced cameras and vision systems, and AI. These technologies enable precise, real-time monitoring and decision-making, enhancing efficiency and reducing waste. Robots may be broadly categorized into five types: collaborative robots, articulated robots, delta robots, cartesian robots, and selective compliance assembly robot arm (SCARA) robots (Riboli et al., 2023). These autonomous systems offer several key advantages in food processing. Robots can handle delicate and complex tasks quickly and skillfully, minimizing damage to plant-based products during processing and packaging. By maintaining high levels of cleanliness and precision, robots help reduce the risk of contamination, thereby preserving food safety and extending shelf life, which directly cuts down on food loss. Robots can operate efficiently in various environments and perform tasks under specific conditions that might be challenging for human workers, further preventing potential waste. (Aly et al., 2023).

Robotics plays a crucial role in transforming plant-based waste and by-products into valuable products. Through precise sorting, separation, and processing, robots can identify and repurpose usable materials that might otherwise be discarded. This not only reduces waste but also creates new revenue streams by converting by-products into ingredients for other food products, animal feed, or bio-based materials, thus contributing to a more sustainable and circular food system (Iñiguez-Moreno et al., 2023; Kshirsagar et al., 2022; Liu et al., 2022; Olawade et al., 2024; Sarc et al., 2019).

For example, Onoda (2020) indicated the potential of robotic sorting systems powered by artificial intelligence algorithms to separate and sort waste, streamline the sorting process, and increase recycling rates. In this process, a stereo camera recognizes images and distances, and the robotic arm automatically sorts them (Ahmed and Asadullah, 2020). Moreover, Soheyb et al. (2021) have proposed a solution based on robotics and IoT technologies for the management of smart greenhouses that offers the farmer the possibility of monitoring, supervising and controlling a large number of greenhouses to minimize waste and damaged production without needing to physically intervene each time for regular actions such as irrigation, fogging and aeration according to the needs of food and plants.

5.7. Blockchain

Blockchain technology serves as a real-time, decentralized digital ledger that securely records and manages transactions across a distributed network. Each transaction, or "block," is linked in a chain that is immutable and transparent, ensuring data integrity and accessibility (Antonucci et al., 2019; Prashar et al., 2020). This technology is particularly valuable in the food supply chain, where it enables the precise tracking and monitoring of food products from farm to table. By utilizing blockchain, data related to the production, transportation, and processing of food can be securely recorded and easily accessed, reducing inefficiencies and mitigating food loss and waste, especially in plant-based products (Carey and Subramanian, 2021; Kör et al., 2022; Wünsche and Fernqvist, 2022).

The ability of blockchain to provide detailed, tamper-proof records ensures that every step in the supply chain is monitored, which helps in identifying and addressing points where food loss occurs. Moreover, the technology facilitates the valorization of food waste by enabling the tracking and management of plant-based by-products, transforming them into valuable secondary products. By ensuring that these by-

products are effectively utilized rather than discarded, blockchain plays a crucial role in promoting sustainability and reducing overall food waste (Gehlot et al., 2022; Li et al., 2023).

6. Conclusions and futures dimensions

The global food system faces risks and threats due to climate change, pandemic, population growth, conflicts, etc., causing massive economic loss. Coincidence of these risks can compound the effects on global food security and nutrition. The European Union's Green Deal is a comprehensive set of policy initiatives aimed at making the EU's economy sustainable and reducing greenhouse gas emissions to net-zero by 2050. It supports a circular economy model, which aims to minimize waste, reuse materials, and recycle products to reduce environmental impact. In the perspective, plant-based waste and by-products are very important because they are valuable resources that may be reused to reduce environmental impact and improve resource efficiency. In addition, the food business may lower greenhouse gas emissions, preserve energy and water, and lessen pollution from landfill disposal by making plant-based waste and by-products valuable. Furthermore, converting these waste streams into high-value products supports the circular economy, which reduces waste and optimizes resource usage. The production of plant-based foods produces nutrient-rich by-products and innovative and emerging technologies are being used to extract beneficial chemicals from these by-products. In addition, recent research has shown that Industry 4.0 technologies such as artificial intelligence, smart sensors, 3D printing, internet of things, and digital twins are being used in the food industry to reduce waste and improve resource management. Operating procedures in the food sector are currently being transformed by digital technology. The use of digital innovations has significant potential to transform the food industry's waste management, resource allocation, and product innovation strategies. In addition to promoting resource recovery through the circular economy, digitalization has created new possibilities to avoid, minimize and recycle waste from specific waste streams in order to provide high-quality secondary raw materials for the waste recycling market and reduce environmental pollution. By embracing digital innovation and adopting circular economy principles, food companies can increase their competitiveness, reduce their environmental footprint and contribute to a more sustainable food system.

However, there are a few issues with Industry 4 technologies. First, there are frequently significant upfront expenditures associated with implementing these technologies. These expenses may be unaffordable, particularly for small and medium-sized businesses. Furthermore, if new technologies cannot be integrated with other systems or are incompatible with the current infrastructure, their implementation may prove to be challenging. Privacy and data security issues also need to be considered. Sensitive corporate data processing and collecting might be risky, thus suitable security precautions must be taken. Ultimately, a trained workforce is necessary for the efficient use of Industry 4.0 technology. However, some firms may find it difficult to acquire and train such competent workers. Notwithstanding these difficulties, based on results of previous research, Industry 4.0 technologies have shown to have a significant impact on achieving sustainability and waste reduction goals. Thus, it is important to implement novel technologies as well as multi-collaboration (such as scientists, food manufacturers, and food technologists) for sustainable waste management in order to achieve circular economy goals and promote sustainable development objectives.

To maximize the impact of reducing food waste, future research should prioritize the integration of Industry 4.0 technologies, such as AI, blockchain, and smart sensors, across all stages of the food supply chain. It is essential to evaluate the cost-effectiveness and scalability of these technologies, considering their application in both developed and developing countries. Additionally, research should focus on adapting these technologies to accommodate various scales of food production

and waste management. Moreover, targeted research on specific plant-based by-products is crucial. Exploring innovative methods for extracting bioactive compounds and investigating their potential health benefits could lead to the development of nutraceuticals and functional foods. Emphasis should be placed on leveraging these by-products to create high-value products that contribute to health and well-being.

Finally, future studies must establish robust sustainability metrics that encompass social, economic, and environmental dimensions. These metrics are vital for assessing the long-term impact of emerging technologies on food sustainability, ensuring that advancements contribute positively to the global food system.

CRediT authorship contribution statement

Abderrahmane Ait-Kaddour: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Conceptualization. **Abdo Hassoun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Inès Tarchi:** Writing – original draft. **Mohammed Loudiyi:** Writing – original draft. **Oumayma Boukria:** Writing – review & editing, Writing – original draft. **Yana Cahyana:** Writing – original draft. **Fatih Ozogul:** Writing – original draft. **Khaoula Khwaldia:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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